

BFO 2020 Generic Dependence Axioms

Concretizes and is concretized by are inverse relations [zba-1]

$$\forall t,a,b (\text{concretizes}(a,b,t) \leftrightarrow \text{isConcretizedBy}(b,a,t))$$

Generically depends on and is carrier of are inverse relations [mvp-1]

$$\forall t,a,b (\text{genericallyDependsOn}(a,b,t) \leftrightarrow \text{isCarrierOf}(b,a,t))$$

Concretizes is dissective on third argument, a temporal region [nyz-1]

$$\forall p,q,r,s (\text{concretizes}(p,q,r) \wedge \text{temporalPartOf}(s,r) \rightarrow \text{concretizes}(p,q,s))$$

Along with qaf 1 this axiom has the consequence that a generically dependent continuant must always be concretized by something [ibk-1]

$$\begin{aligned} \forall t,g (\text{instanceOf}(g,\text{genericallyDependentContinuant},t) \\ \rightarrow \exists s,tp (\text{temporalPartOf}(tp,t) \wedge \text{concretizes}(s,g,tp))) \end{aligned}$$

A g dependent continuant b g depends on an independent continuant c at t means: there inheres in c at t an s dependent continuant which concretizes b at t [otx-1]

$$\begin{aligned} \forall g,c,t (\text{genericallyDependsOn}(g,c,t) \\ \rightarrow \exists s,tp (\text{temporalPartOf}(tp,t) \wedge \text{inheresIn}(s,c) \wedge \text{concretizes}(s,g,tp))) \end{aligned}$$

Concretizes is time indexed and has domain: specifically dependent continuant or process and range: generically dependent continuant [rog-1]

$$\begin{aligned} \forall a,b,t (\text{concretizes}(a,b,t) \\ \rightarrow (\text{instanceOf}(a,\text{specificallyDependentContinuant},t) \vee \text{instanceOf}(a,\text{process},t)) \\ \wedge \text{instanceOf}(b,\text{genericallyDependentContinuant},t) \\ \wedge \text{instanceOf}(t,\text{temporalRegion},t)) \end{aligned}$$

Generically depends on is time indexed and has domain: generically dependent continuant and range: independent continuant but not spatial region [ekp-1]

$$\begin{aligned} \forall a,b,t (\text{genericallyDependsOn}(a,b,t) \\ \rightarrow \text{instanceOf}(a,\text{genericallyDependentContinuant},t) \\ \wedge \text{instanceOf}(b,\text{independentContinuant},t) \wedge \neg \text{instanceOf}(b,\text{spatialRegion},t) \\ \wedge \text{instanceOf}(t,\text{temporalRegion},t)) \end{aligned}$$

If a specifically dependent continuant concretizes a gdc then the gdc generically depends on the bearer of the sdc [cik-1]

$$\begin{aligned} \forall g,b,sdc (\exists t \text{instanceOf}(g,\text{genericallyDependentContinuant},t) \\ \wedge \exists t \text{instanceOf}(sdc,\text{specificallyDependentContinuant},t) \\ \wedge \exists t \text{instanceOf}(b,\text{independentContinuant},t) \\ \rightarrow \forall t (\text{concretizes}(sdc,g,t) \wedge \text{inheresIn}(sdc,b) \rightarrow \text{genericallyDependsOn}(g,b,t))) \end{aligned}$$

If a generically dependent continuant participates in a process p then, if it is concretized as a process, that process is part of p, and if concretized as an sdc then the bearer of that sdc participates in the process [fmm-1]

$$\begin{aligned} \forall gdc,p,t (\text{instanceOf}(gdc,\text{genericallyDependentContinuant},t) \wedge \text{participatesIn}(gdc,p,t) \\ \rightarrow \exists tp,b (\text{temporalPartOf}(tp,t) \wedge \text{concretizes}(b,gdc,tp) \\ \wedge ((\text{instanceOf}(b,\text{specificallyDependentContinuant},tp) \\ \wedge (\exists ic (\text{specificallyDependsOn}(b,ic) \wedge \text{participatesIn}(ic,p,tp)))) \\ \vee (\text{occurrentPartOf}(b,p) \wedge \text{existsAt}(b,tp)))))) \end{aligned}$$

